140

Cutaneous Continent Urinary Diversion

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Keywords

cecum; cutaneous continent urinary diversion; terminal ileum

Questions

1. A 45-year-old man had an ileal conduit diversion as a child for bladder exstrophy. He requests a continent diversion. Serum creatinine is 2 mg/dL. Loopogram shows bilaterally thin ureters with small kidneys. Which is the best procedure?
    a. Ureterosigmoidostomy
    b. T pouch using the ileal conduit
    c. Abandon continent diversion
    d. Penn pouch using the ileal conduit
    e. Indiana pouch
2. A 45-year-old man underwent ileal conduit urinary diversion as a child for bladder exstrophy. He presents requesting continent diversion. Serum creatinine is 2 mg/dL. Loopogram shows bilateral hydronephrosis and a pipestem conduit. What is the best course of action?
    a. Mainz II to avoid problems with dilated ureters
    b. T pouch abandoning the disease conduit
    c. No continent diversion

d. Drain the upper tracts and reassess renal function
e. Proceed to neobladder construction

3. A patient undergoing a cystectomy and planned continent cutaneous diversion has positive ureteral margin biopsies up to 2 cm above each iliac artery, at which point negative biopsies are obtained. What is the best course of action?
    a. Use the terminal ileum for ureteral implantation and a Mitrofanoff continence mechanism
    b. No continent diversion
    c. Mobilize the kidneys and stretch the ureters to the reservoir
    d. Use a T pouch with a long chimney
    e. Cutaneous ureterostomies
4. Preservation of the ileocecal valve can be maintained with which catheterizable pouch?
    a. T pouch or Kock pouch
    b. Le Bag
    c. Indiana pouch
    d. Mainz I or II
    e. Penn pouch
5. In which procedure to repair a nipple valve would resection of additional bowel be routinely required?
    a. Stones on exposed staples
    b. Nipple valve slippage
    c. Nipple valve atrophy
    d. Pinhole leak
    e. Anastomotic leak
6. A 10-year-old child has an ileal conduit for myelomeningocele. The conduit was replaced on two occasions for pipestem conduit development. The conduit is again affected by the same process. The patient's family wants a continent diversion. Which is the best procedure?
    a. Ureterosigmoidostomy
    b. Revise the conduit
    c. T pouch using the ileal conduit
    d. Penn pouch using the ileal conduit
    e. Indiana pouch using the ileal conduit

7. A patient with chronic active hepatitis and invasive bladder cancer associated with intravesical carcinoma in situ is scheduled for a cystoprostatectomy. The serum creatinine concentration is 1 mg/dL. Prostatic urethral biopsy shows mild atypia. What is the best diversion?
    a. T pouch
    b. Ileal conduit
    c. Right colon reservoir
    d. Mainz II
    e. Cutaneous ureterostomies
8. The highest reoperation rate in catheterizable pouches occurs with what type of sphincter?
    a. In situ appendix
    b. Imbricated terminal ileum
    c. Plicated terminal ileum
    d. Nipple valves
    e. Transposed appendix
9. Which of the Mitrofanoff sphincter deficiencies can be corrected surgically?
    a. Length of the appendix
    b. Absence of the appendix
    c. Stenosis of the appendix
    d. All of the above
10. Hematuria and skin breakdown may occur with what type of pouch?
    a. T
    b. Gastric
    c. Mainz
    d. Right colon
    e. All of the above
11. Preoperative colonoscopy is indicated in candidates for which reservoir procedures?
    a. Ileal
    b. Jejunal
    c. Rectal
    d. Gastric

e. All of the above

12. What condition is more common in absorbable stapled ileal pouches?
    a. Urine leaks
    b. Valve failure
    c. Hydronephrosis
    d. Ischemic pouch contraction
    e. Ureteral stricture
13. Anastomotic transitional cell carcinoma develops in a patient who has undergone cystectomy and continent cutaneous urinary diversion. What is the best treatment?
    a. Distal ureterectomy and reimplantation
    b. Conversion to ileal conduit
    c. Ileal ureter interposition
    d. Nephroureterectomy
    e. Cutaneous ureterostomies
14. Drainage of mucus is most difficult with which sphincteric mechanism?
    a. Kock valve
    b. In situ appendix
    c. Imbricated ileum
    d. Plicated ileum
    e. Transposed appendix
15. Which continent cutaneous diversion allows for a refluxing ureteroenteric anastomosis?
    a. Mitrofanoff with implantation of the ureters into terminal ileum
    b. Mitrofanoff with implantation of the ureters into the colon
    c. T pouch
    d. Kock pouch
    e. Indiana pouch
16. Three years after radical cystectomy and construction of a Kock pouch, a patient presents with right lower quadrant discomfort and associated spurts of urinary leakage. The test most likely to diagnose the condition is:
    a. computed tomography (CT).

b. intravenous pyelogram (IVP).
c. urine culture and sensitivity.
d. cystogram of pouch.
e. urodynamics.

17. Three years after cystectomy and Kock pouch for bladder cancer, a patient presents with recurrent episodes of bilateral pyelonephritis. The test most likely to provide the correct diagnosis is:
    a. CT.
    b. IVP.
    c. urine culture and sensitivity.
    d. cystogram of the pouch.
    e. magnetic resonance imaging (MRI).
18. What is the most important feature in preventing nipple valve slippage?
    a. Absorbable staples
    b. Length of the intussusception
    c. Resecting adequate mesentery
    d. Attaching the nipple valve to the side wall of the reservoir
    e. Length of staple line
19. In a patient with pipestem conduit and bilateral hydronephrosis requesting conversion to continent urinary diversion, nephrostomy drainage results in clearance values of 40 mL/min on the right and 10 mL/min on the left. Serum creatinine is 1.8 mg/dL. The next step in management is:
    a. Mainz II to avoid problems with the dilated ureters.
    b. T pouch abandoning the disease conduit.
    c. no continent diversion.
    d. ureterosigmoidostomy.
    e. neobladder.
20. A patient with squamous cell cancer of the bladder desires cystectomy and continent diversion. He has lost 20 pounds in the month before surgery. The next step in management is:
    a. increased oral intake.
    b. conduct preoperative hyperalimentation.
    c. conduct postoperative hyperalimentation.
    d. proceed directly with surgery.

e. count calories.

21. Preoperative evaluation with an oatmeal enema is required in which procedure?
    a. Right colon reservoir
    b. Mainz I pouch
    c. Mainz II procedure
    d. Le Bag pouch
    e. Indiana pouch
22. Follow-up urinary cytology and colonoscopy should be used in which type of continent diversion?
    a. Ureterosigmoidostomy
    b. Mainz II procedure
    c. Right colon reservoir
    d. All of the above
23. Nocturnal emptying of the patient's reservoir is required in which type of diversion?
    a. Ureterosigmoidostomy
    b. T pouch
    c. Right colon reservoir
    d. Penn pouch
    e. Ileal conduit
24. The appendix is sacrificed in patients undergoing which pouch construction?
    a. Indiana
    b. Le Bag
    c. Mainz I
    d. All of the above
25. Pouch stone development occurs most commonly with which pouch?
    a. T pouch
    b. Kock pouch
    c. Penn pouch
    d. Gastric-ileal composite pouch
    e. Le Bag
26. What is the typical catheter used for appendiceal sphincters?
    a. 22-French (Fr) straight-tipped

b. 22-Fr coudé-tipped
c. 14-Fr straight-tipped
d. 14-Fr coudé-tipped
e. 20-Fr coudé-tipped

27. Urinary retention resulting from continent diversion occurs most commonly with what type of sphincter?
    a. Appendiceal stoma
    b. Benchekroun hydraulic valve
    c. Nipple valve sphincter
    d. Imbricated Indiana mechanism
28. Immediate postoperative initial pouch capacity is least in which pouch?
    a. T or Kock ileal
    b. Right colon
    c. Gastric
    d. Mainz I
    e. Transverse colon
29. Elevated pouch pressures would potentially facilitate the continence mechanism seen with which valve or sphincter?
    a. Benchekroun ileal valve
    b. Kock valve
    c. Appendiceal tunnel
    d. Imbricated Indiana mechanism
    e. All of the above
30. The long-term failure rate of continence mechanisms is greatest with which mechanism?
    a. T pouch valve
    b. Appendiceal tunnel
    c. Benchekroun hydraulic valve
    d. Imbricated terminal ileum
31. Absorbable staples in continent urinary diversion are best suited to what type of reservoir pouch?
    a. Ileal
    b. Right colon reservoir
    c. Gastric-ileal composite
    d. Gastric

e. None of the above

32. When creating a large intestinal reservoir from absorbable staples, why is bowel eversion necessary?
    a. Because staples should not be used in reservoir construction
    b. To inspect the inside of the reservoir
    c. To avoid injury to the mesenteric blood supply
    d. To allow application of the second row of staples
    e. None of the above
33. Which of the following conditions make patients unsuitable candidates for continent urinary diversion?
    a. Multiple sclerosis
    b. Quadriplegia
    c. Mental impairment
    d. Severe physical impairment
    e. All of the above
34. Which of the following sutures should NOT be used in the construction of a reservoir?
    a. Chromic catgut
    b. Plain catgut
    c. Silk
    d. Polyglycolic acid (Dexon)
    e. Polyglactin (Vicryl)
35. Which of the following diversions place the patient at risk for the development of a late malignancy?
    a. Ureterosigmoidostomy
    b. T pouch
    c. Mainz II
    d. Indiana pouch
    e. All of the above
36. Which of the following diversions places the patient at greatest risk for the development of a late malignancy?
    a. Ureterosigmoidostomy
    b. T pouch
    c. Mainz II
    d. Indiana reservoir
    e. Le Bag

37. Continent urinary diversion has which of the following effects?
    a. Results in a psychotic depression
    b. Results in an improved psychosocial adjustment
    c. Results in violent behavior
    d. Bipolar behavior
    e. None of the above
38. According to most randomized studies, which type of urinary diversion is associated with the highest reported quality of life?
    a. Ureterosigmoidostomy
    b. Continent ileal reservoir (Kock pouch)
    c. Ileal conduit
    d. Orthotopic neobladder
    e. None—no conclusive studies have established higher satisfaction or quality of life with any one specific continent diversion
39. Which of the following is NOT true of continent urinary diversion?
    a. It is the gold standard of urinary diversion.
    b. It is a safe and reliable urinary diversion.
    c. It is associated with an increased complication rate.
    d. It is appropriate for selected individuals.
    e. It requires stricter selection criteria than incontinent diversion.
40. Which of the following circumstances would contraindicate a rectal bladder?
    a. Prior pelvic irradiation
    b. Unilateral ureteral dilation
    c. Bilateral ureteral dilation
    d. Lax anal sphincter tone
    e. All of the above
41. During the construction of a continent cutaneous urinary diversion, the surgeon should:
    a. not be concerned about the continence mechanism because the mechanism will mold to the catheter.
    b. not test the continence mechanism for ease of catheterization.
    c. not be concerned about pouch integrity because the pouch will seal itself.
    d. do none of the above.

e. do all of the above.

42. If the urine in a continent cutaneous reservoir is found to be infected, what should be done?
    a. Nothing needs to be done in the absence of symptoms.
    b. The urine should always be sterilized with appropriate antibiotics.
    c. The infection should be eradicated, and prophylactic antibiotics prescribed.
    d. Administer an intravenous pyelogram to check for upper tract damage.
    e. Perform a pouch-o-gram.
43. The most appropriate and conservative care for pouch rupture is:
    a. broad-spectrum antibiotic therapy.
    b. careful radiologic imaging and antibiotic therapy.
    c. surgical exploration for repair of the rupture and broad-spectrum antibiotic therapy.
    d. pouch drainage and broad-spectrum antibiotic therapy.
    e. bilateral percutaneous nephrostomies.
44. The first pouch to use the Mitrofanoff principle was the:
    a. Mainz I.
    b. Penn.
    c. Kock.
    d. Indiana.
    e. Le Bag.
45. Which of the following represents the advantage of the gastric pouch?
    a. Electrolyte reabsorption is reduced.
    b. Absorptive malabsorption is avoided.
    c. Acid urine may reduce the risk of infection.
    d. All of the above
    e. None of the above
46. When converting from an ileal conduit to a continent diversion, the conduit should be:
    a. discarded because it is older and subject to higher complications.
    b. preserved for the ureteroileal anastomosis.

c. incorporated into the continent diversion when possible.
d. discarded because it is a potential nidus of infection.
e. None of the above

47. Which of the following is TRUE of absorbable staples?
    a. Their use has been shown to shorten operative time.
    b. They are safe and reliable.
    c. Unlike nonabsorbable staples, they must not be overlapped.
    d. All of the above.
    e. None of the above.

Answers

1. **c. Abandon continent diversion.** A creatinine level greater than 1.8 mg/dL indicates a level of renal function insufficient for continent diversion.
2. **d. Drain the upper tracts and reassess renal function.** The best course of action is to place ureteral cutaneous stents bilaterally (bypassing the pipestem segment) and reassess urinary function. In evaluating the hydronephrotic patient with impaired renal function for continent diversion, upper tract drainage is advised. If necessary, bilateral nephrostomy tubes can be used.
3. **a. Use the terminal ileum for ureteral implantation and a Mitrofanoff continence mechanism.** The best course of action is to perform a right colon reservoir with anastomosis of the ureters to the terminal ileum. The appendix or other pseudo-appendiceal (Mitrofanoff) mechanisms can be used for continence. The terminal ileum can accommodate short ureters.
4. **a. T pouch or Kock pouch.** Preservation of the ileocecal valve can be maintained with the T or Kock pouch. All other pouches use the right colon, so that the ileocecal valve is sacrificed.
5. **c. Nipple valve atrophy.** Nipple valve atrophy requires that a new nipple valve be made of additional bowel.
6. **b. Revise the conduit. With significant small bowel compromise, as well as loss of the ileocecal valve in a neurogenic bladder patient, severe diarrhea may ensue.**

7. **b. Ileal conduit.** The best approach is cystoprostatectomy and a conduit. Normal hepatic function is mandated in any patient undergoing continent diversion.
8. **d. Nipple valves.** The highest reoperation rate is associated with nipple valve sphincter failure.
9. **d. All of the above.** The caliber of Mitrofanoff mechanisms, the length of the appendix, stenosis, and even absence of the appendix can be resolved by surgical variations.
10. **b. Gastric.** Hematuria and cutaneous skin erosion may occur with a gastric pouch. With gastric reservoirs or composite reservoirs, the low pH of the urine may lead to hematuria and cutaneous breakdown.
11. **c. Rectal.** Preoperative colonoscopy is relatively indicated in candidates for any pouch.**Any pouch using colon mandates preoperative colonic evaluation.**
12. **d. Ischemic pouch contraction. Because of the overlap of staple lines in absorbable stapled ileal pouches, ischemic pouch contraction may occur.**
13. **a. Distal ureterectomy and reimplantation.** An additional segment of ileum can serve as a proximal limb to the reservoir. If nephrectomy is necessary, careful attention must be paid to the residual renal function.
14. **b. In situ appendix. The small-diameter catheter used in draining appendiceal sphincter pouches allows for less effective mucus drainage.**
15. **a. Mitrofanoff with implantation of the ureters into terminal ileum.** The implantation of the ureters into the terminal ileum may allow for reflux. The ileocecal valve and the isoperistaltic ileal segment may either prevent or diminish reflux.
16. **c. Urine culture and sensitivity.** The most important diagnostic test is urine culture. The symptoms described are those of pouchitis. This is treated by appropriate antibiotic therapy.
17. **d. Cystogram of the pouch.** The proximal nipple valve may have failed, leading to reflux and pyelonephritis. This is tested by the pouch-o-gram.

18. **d. Attaching the nipple valve to the side wall of the reservoir.** This results in a relative lengthening of the valve rather than a foreshortening of the valve with pouch filling.
19. **c. No continent diversion. In this case, although the serum creatinine level returns to 1.8 mg/dL, the clearance value measured is less than the 60 mL/min required for continent diversion. Continent diversion should be abandoned, and simple replacement of the conduit considered.**
20. **b. Conduct preoperative hyperalimentation.** The 20-pound weight loss indicates a potential for nutritional depletion or metastatic disease. A careful search for metastatic disease should be undertaken. For the patient with nutritional depletion, preoperative hyperalimentation is suggested to be of value.
21. **c. Mainz II procedure. Any procedure that relies on the intact anal sphincter for continence (i.e., the Mainz II pouch) requires an assessment of the sphincter before carrying out the operation. This can be assessed by an oatmeal enema, which mimics the constitution of a combination of the urinary and fecal streams.**
22. **d. All of the above. Follow-up urinary cytology and colonoscopy is mandatory with any procedure that combines urinary and fecal streams. Because of an increased risk of malignancy even in the absence of admixture of urine and stool, all large intestinal pouches should be subjected to annual investigation by pouchoscopy and cytology.**
23. **a. Ureterosigmoidostomy. Nocturnal reservoir emptying may be required with any of the continent cutaneous reservoirs to prevent overdistention and possible rupture but is mandatory with ureterosigmoidostomy owing to the additional risk of fecal incontinence and metabolic acidosis.**
24. **d. All of the above.** The appendix is sacrificed in patients undergoing Indiana, Le Bag, and Mainz I pouch reconstruction because it can serve as a nidus for infection and abscess formation.
25. **b. Kock pouch.** Pouch stone development occurs most commonly with the Kock pouch. Despite the exclusion of distal staples, the stapling techniques used to secure nipple valves will lead to a

higher potential for stone development than in pouches not requiring nipple valves.

26. **d. 14-Fr coudé-tipped.** Larger catheters will not fit into the appendix. A straight catheter is more difficult to pass.
27. **c. Nipple valve sphincter. Urinary retention occurs most commonly with nipple valve sphincters.** If the chimney of the nipple valve is not near the surface of the abdomen, the catheter can be misdirected into folds of bowel rather than through the nipple valve.
28. **a. T or Kock ileal.** Immediate postoperative initial pouch capacity is least in ileal reservoirs (i.e., the T or Kock pouch).**Small bowel pouches have initial capacities that are much lower than right colon pouches.**
29. **a. Benchekroun ileal valve.** Because the Benchekroun ileal valve is hydraulic, higher pouch pressures would facilitate continence, whereas lower pouch pressures might lead to incontinence.
30. **c. Benchekroun hydraulic valve.** The long-term outcome of Benchekroun hydraulic ileal valve mechanisms is possibly the worst of all reported sphincteric mechanisms.
31. **b. Right colon reservoir. The use of absorbable staples is best suited to large bowel pouches. With large bowel pouches there is no problem with staple lines causing subsequent bowel ischemia.**
32. **d. To allow application of the second row of staples.** In an absorbable-stapled right colon pouch, bowel eversion is required to allow for the application of the second row of staples. **Staple lines must not cross because this will prevent the bulky, absorbable staples from seating properly.** The bowel is everted, a cut is made beyond the end of the staple line, and the next line of staples is applied.
33. **e. All of the above. Patients with multiple sclerosis, quadriplegia, frailty, or mental impairment will at some point in their lives require the care of family members or visiting nurses, so they are poor candidates for any form of continent diversion.**
34. **c. Silk.** All sutures used in the urinary tract should be absorbable.

35. **e. All of the above.** Late malignancy has been reported in all bowel segments exposed to the urinary stream, whether or not there is a commingling with feces.
36. **a. Ureterosigmoidostomy. Although late malignancy has been reported in all bowel segments exposed to the urinary stream, whether or not there is a commingling with feces, the mixture of urothelium, urine, and feces poses the greatest risk.**
37. **b. Results in an improved psychosocial adjustment.** Many studies from throughout the world have suggested an improved psychosocial adjustment of the patient undergoing continent urinary and fecal diversion compared with those patients with diversions requiring collecting appliances.
38. **e. None—no conclusive studies have established higher satisfaction or quality of life with any one specific continent diversion. There are insufficient quality-of-life data from randomized studies comparing continent and incontinent urinary diversions to establish the superiority of any one technique.**
39. **a. It is the gold standard of urinary diversion.** Ileal conduit should be considered the "gold standard" of urinary diversion.
40. **e. All of the above.** Dilated ureters, pelvic irradiation, and lax anal sphincteric tone are all contraindications to the procedure.
41. **d. Do none of the above. The continence mechanism must be catheterized intraoperatively to ensure ease of catheter passage.** This is an extremely important and crucial maneuver because the inability to catheterize is a serious complication that will often result in the need for reoperation.
42. **a. Nothing needs to be done in the absence of symptoms.** Most authors would suggest that bacteriuria in the absence of symptomatology does not warrant antibiotic treatment.
43. **c. Surgical exploration for repair of the rupture and broad-spectrum antibiotic therapy. In general, these patients require immediate pouch decompression, radiologic pouch studies, and surgical exploration with pouch repair.** If the amount of urinary extravasation is small and the patient does not have a surgical abdomen, catheter drainage and antibiotic administration may

suffice in treating intraperitoneal rupture of a pouch. Patients managed with this conservative approach require careful monitoring.

44. **b. Penn.** The Penn pouch was the first continent diversion to use the Mitrofanoff principle, wherein the appendix served as the continence mechanism.
45. **d. All of the above.** Electrolyte reabsorption is greatly diminished, shortening of the absorptive bowel does not occur, and the acid urine may decrease the likelihood of reservoir colonization.
46. **c. Incorporated into the continent diversion when possible.** The authors prefer to use the conduit in some form whenever possible. The use of an existing bowel segment has the potential to diminish metabolic sequelae and may result in a lower complication rate.
47. **d. All of the above.** The use of absorbable staples has substantially reduced the time required to fashion bowel reservoirs and has demonstrated short-term and long-term reliability with respect to reservoir integrity and volume. They must not be overlapped because overlapping will prevent the proper close of the staple.

Chapter Review

1. The ability to self-catheterize is essential in patients who are to be considered for a continent cutaneous diversion.
2. All patients should be prepared for the possibility of a traditional ileal conduit if intraoperative circumstances warrant it.
3. A patient should have a minimum creatinine clearance of 60 mL/min to undergo a continent urinary diversion.
4. Single J ureteral stents are used in all continent diversions. The stents are brought out through a separate abdominal stab wound, and a Malecot catheter should be placed into the reservoir and brought out through a separate stab wound as well.
5. In continent diversions, it is not clear at this time whether antirefluxing ureteral intestinal anastomoses are necessary to preserve the upper tracts; however, antirefluxing procedures are associated with a higher incidence of stricture over the long term.

6. Most patients are satisfied with the type of urinary diversion, irrespective of whether it is continent or not.
7. It is often useful to secure the reservoir to the anterior abdominal wall to prevent the reservoir from migrating. This is conveniently done where the Malecot exits the reservoir onto the anterior abdominal wall.
8. Renal and hepatic function must be carefully evaluated before a continent diversion is performed. Significant abnormalities in either are a contraindication to continent diversion. The glomerular filtration rate should be 60 mL/min or greater.
9. Patients with rectal bladders are very prone to the complication of hyperchloremic acidosis and total body potassium depletion. These patients also have an increased incidence of rectal cancer.
10. The loss of the ileocecal valve in patients with neurologic or intestinal disorders subjects the patient to a significant risk of debilitating diarrhea.
11. Any procedure that relies on the intact anal sphincter for continence (i.e., the Mainz II pouch) requires an assessment of the sphincter before carrying out the operation. This can be assessed by an oatmeal enema.
12. Because of an increased risk of malignancy even in the absence of admixture of urine and stool, all large intestinal pouches should be subjected to annual investigation by pouchoscopy and cytology.
13. Nocturnal reservoir emptying may be required with any of the continent cutaneous reservoirs to prevent overdistention and possible rupture, but it is mandatory with ureterosigmoidostomy because of the additional risk of fecal incontinence and metabolic acidosis.
14. Small bowel pouches have initial capacities that are much lower than those of right colon pouches.
15. The use of absorbable staples is best suited to large bowel pouches. With large bowel pouches there is no problem with staple lines causing subsequent bowel ischemia.
16. Although late malignancy has been reported in all bowel segments exposed to the urinary stream, whether or not there is a commingling with feces, the juxtaposition of urothelium, urine, and feces poses the greatest risk.

141

Orthotopic Urinary Diversion

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Keywords

bladder cancer; continent urinary diversion; cystectomy; orthotopic diversion

Questions

1. Which of the following was the key finding that allowed application of orthotopic urinary diversion to women undergoing cystectomy?
    a. Confirmation that an intact bladder neck is required for continence
    b. Demonstration in cystectomy specimens that urethral involvement was rare in the absence of tumor at the bladder neck
    c. Understanding of the relationship between estrogen levels and continence in elderly women
    d. Studies showing that direct invasion into the uterus is relatively rare in women with invasive bladder cancer
    e. Quality of life studies showing that men with continent diversion had better quality of life than those with ileal conduit
2. The risk factor most predictive for urethra recurrence following cystectomy for urothelial carcinoma is:
    a. prostatic stromal invasion.
    b. node-positive disease.

c. carcinoma-in-situ in females.
d. pathologic stage pT3b tumor at the trigone.
e. history of multiple prior tumors.

3. A 80-year-old man with clinical cT2 bladder cancer lives alone but is active. His serum creatinine is 1.0 following neoadjuvant gemcitabine and cisplatin chemotherapy. He is interested in an orthotopic diversion. Which of the following would you tell him about that option?
    a. Neoadjuvant chemotherapy increases the early complications of orthotopic diversion.
    b. Older patients take longer to regain continence than younger patients.
    c. Ileal conduit will be easier for him to take care of than a continent diversion.
    d. Continent diversion is contraindicated in men over 75 years old.
    e. His risk of renal deterioration with continent diversion is higher than with ileal conduit.
4. Which of the following patients should NOT be offered an orthotopic neobladder?
    a. An 82-year-old healthy woman with recurrent cT1 and CIS following intravesical BCG and a prior vaginal hysterectomy
    b. A 53-year-old woman with an eGFR of 55 following neoadjuvant chemotherapy
    c. A 50-year-old man 2 years following low anterior colon resection with adjuvant chemotherapy and external beam radiation to the pelvis
    d. A 60-year-old woman with diabetes and hypertension
    e. A 58-year-old woman with palpable induration of the anterior vaginal apex
5. Which of these is a key requirement for construction of an orthotopic diversion?
    a. It should prevent vesicoureteral reflux to preserve renal function.
    b. It should be made of ileum or a combination of colon and ileum.

c. The bowel used should be detubularized and fashioned into a spherical shape.
d. It should be made with the smallest amount of bowel possible.
e. The ureters should be plugged into an isoperistaltic segment of bowel.

6. Which of the following have been suggested to decrease the risk of urinary retention following ileal neobladder in women?
   a. Regular urethral dilation
   b. Tack the pouch to the anterior abdominal wall
   c. Biofeedback training in the early postop period
   d. Preservation of the uterus
   e. Construct a W pouch rather than a Studer type pouch
7. In performing a cystectomy and orthotopic ileal neobladder in a male, the most important step in preserving continence is to:
   a. construct a large capacity reservoir.
   b. avoid excess dissection anterior to the urethra.
   c. perform a nerve-sparing procedure in all cases.
   d. avoid removal of the presacral lymph nodes.
   e. place a suprapubic catheter during the early postoperative period.
8. A 64-year-old man with recurrent CIS who strongly prefers an orthotopic or continent cutaneous diversion is found to have grossly node-positive disease at surgery. The next step is:
   a. close and refer for chemotherapy and radiation.
   b. complete the cystectomy but do an ileal conduit.
   c. complete the cystectomy but do a continent cutaneous diversion.
   d. complete the cystectomy and neobladder and refer for adjuvant chemotherapy.
   e. complete the cystectomy and neobladder and refer for adjuvant radiation therapy.
9. Prior to considering a continent orthotopic diversion, what evaluation is mandatory?
   a. Prostatic urethral biopsy
   b. Evaluation of renal function
   c. Colonoscopy to rule out colon polyps

d. Biopsy of the bladder neck in a female
e. Video urodynamics to test the integrity of the external sphincter

10. The primary innervation of the rhabdosphincter responsible for continence in men and women following an orthotopic diversion is:
    a. parasympathetics from S2 to S4.
    b. anterior branches of the sciatic nerve.
    c. sympathetic nerves from the hypogastric plexus.
    d. pudendal nerve.
    e. femoral nerve.
11. Use of metallic surgical staples should be avoided in construction of a continent diversion because:
    a. it is less secure than a hand-sewn closure.
    b. they tend to be buried in the bowel mucosa.
    c. the staples increases the risk of subsequent infection.
    d. the staples can become a nidus for stone formation.
    e. they increase the risk of cancer developing in the segment.
12. A 71-year-old male is found on routine follow-up to have a pelvic recurrence 13 months after cystectomy and ileal neobladder. The mass is 2.5 cm in the obturator fossa, abutting the pouch. There is no hydronephrosis. He has good daytime continence but occasionally leaks at night. The next step is:
    a. resection of the mass with removal of the pouch and conversion to an ileal conduit.
    b. cystoscopy to look for invasion of the reservoir.
    c. placement of a permanent suprapubic tube.
    d. resection of the mass with preservation of the neobladder.
    e. systemic chemotherapy with or without external beam radiation.
13. Asymptomatic bacteruria in patients with orthotopic diversion:
    a. carries a high risk of subsequent pyelonephritis.
    b. leads to an increase in urethral recurrence.
    c. does not generally require treatment.
    d. is very rare in most reported series.
    e. suggests probable outlet obstruction.

14. A 59-year-old man is 6 years out from a radical cystectomy and neobladder. He had excellent day and nighttime continence and no problems with infections, but recently has started to leak at night. The next step is:
    a. fluorourodynamics.
    b. CT scan looking for local recurrence.
    c. check postvoid residual.
    d. trial of long-term antibiotics.
    e. MRI of the spine.
15. A 66-year-old male 2 years after a cystectomy and Hautmann ileal neobladder for pathologic stage T2N0M0 bladder cancer is found on routine CT scan to have a very distended neobladder and mild bilateral hydronephrosis. He has a postvoid residual of over 800cc. Cystoscopy and digital rectal exam are normal. The next step is:
    a. teach the patient intermittent catheterization.
    b. dilate the urethra with van Buren sounds.
    c. instruct the patient to Crede while Valsalva voiding.
    d. convert the diversion to an ileal conduit
    e. decompress the neobladder with a catheter for 2 weeks and then resume regular voiding.
16. Quality-of-life studies of patients with orthotopic diversion
    a. are best done by the physician asking the patient about the function of his/her neobladder.
    b. have consistently shown that patients with continent diversions have a better quality of life than those with ileal conduits.
    c. can be easily done with currently available questionnaires used for other populations.
    d. have often been underpowered or affected by selection bias.
    e. have shown that most patients with any urinary diversion have very poor quality of life.
17. A 70-year-old man who is 10 days s/p cystectomy and neobladder is readmitted with fever, and CT scan shows a large fluid collection near the reservoir that fills with contrast on delayed images. The catheter and ureteral stents are still in place. The next step is:

a. IV antibiotics and observation with frequent catheter irrigation.
b. exploration and repair of the pouch.
c. bilateral percutaneous nephrostomy tube placement.
d. percutaneous drainage of the fluid collection.
e. percutaneous placement of a suprapubic catheter.

18. A 53-year-old woman had an anterior exenteration and neobladder with omental flap interposition 3 months previously. She still has total incontinence day and night. The next step is:
    a. reassurance and reinforce Kegel exercises.
    b. refer for physical therapy for pelvic floor strengthening.
    c. evaluate for possible vesicovaginal fistula.
    d. prescribe extended-release oxybutynin.
    e. fluorourodynamics.
19. A 70-year-old woman had a cystectomy and ileal neobladder diversion 5 years ago for pT2N0 urothelial cancer. CT scan is normal, and she has excellent continence and empties well with negative urine culture. The next step is:
    a. refer her to her primary care physician for routine health maintenance.
    b. continue annual CT abdomen and pelvis out to 10 years.
    c. annual endoscopy of the pouch to screen for secondary malignancy.
    d. annual cystogram and serum creatinine to evaluate for reflux nephropathy.
    e. renal ultrasound, chemistry panel, and vitamin B12 level every 1 to 2 years.
20. Which of the following is NOT usually part of an Early Recovery After Surgery (ERAS) protocol applied to patients undergoing radical cystectomy?
    a. Alvimopan BID beginning AM of surgery
    b. Early feeding
    c. No mechanical or antibiotic bowel prep
    d. Continuous IV narcotics to optimize pain management
    e. Early removal of nasogastric tubes

21. An otherwise healthy 90-year-old man had a cystectomy and ileal neobladder 15 years previously with a hemi-Kock pouch to the urethra. He presented with a creatinine of 4.0 and bilateral hydronephrosis on renal ultrasound that did not resolve with catheter drainage. The most likely cause of his problem is:
    a. bilateral ureteral stones.
    b. stenosis of the afferent nipple valve.
    c. reflux nephropathy.
    d. urinary retention.
    e. cancer recurrence in the reservoir.
22. Which of the following is true about robotic cystectomy with extracorporeal neobladder construction compared with standard open cystectomy?
    a. Patients undergoing robotic-assisted laparoscopic radical cystectomy (RARC) have significantly fewer early and late complications.
    b. Continence has been shown to be improved.
    c. Long-term cancer control has been proven to be equivalent.
    d. The surgery can be performed through a smaller incision with lower blood loss.
    e. Hospital stay is consistently shorter.
23. A 66-year-old man had a cystectomy and sigmoid neobladder 4 years previously and has good continence. He was noted on recent CT scan to have two 0.7 cm calcifications in the pouch. The next step is:
    a. reassurance that the stones will probably pass.
    b. shock wave lithotripsy.
    c. metabolic stone evaluation.
    d. begin intermittent catheterization
    e. cystoscopy and laser lithotripsy of the stone.
24. A 49-year-old man had a cystectomy and neobladder 4 months previously. He has excellent continence in the daytime but still has accidents at night even though he gets up twice to empty. The next step is:
    a. reassurance that the nighttime continence will likely improve with more time.

b. trial of extended release oxybutynin.
c. strict fluid restriction to no more than 1500cc per day.
d. intermittent catheterization.
e. refer for physical therapy for pelvic floor strengthening.

Answers

1. **b. Demonstration in cystectomy specimens that urethral involvement was rare in the absence of tumor at the bladder neck.** The two findings that paved the way for orthotopic diversion in women were retrospective pathologic studies showing that urethral involvement was rare in the absence of bladder neck involvement (in other words, “skip lesions” were rarely seen, suggesting oncologic safety), and studies showing that women could be continent without an intact bladder neck (previously thought to be required).
2. **a. Prostatic stromal invasion.** Two large studies have demonstrated conclusively that urethral recurrence in men is associated with prostatic stromal invasion in the cystectomy specimen.
3. **b. Older patients take longer to regain continence than younger patients.** Older men regain continence more slowly than younger men, but the majority of fit older men will ultimately have good control, especially in the daytime. Older men often have difficulty becoming independent in managing an ileal conduit, and in a man living alone a neobladder may actually be simpler.
4. **A 58-year-old woman with palpable induration of the anterior vaginal apex.** Women with palpable invasion of the anterior vaginal wall have a high risk of urethral tumor and should not undergo orthotopic diversion.
5. **c. The bowel used should be detubularized and fashioned into a spherical shape.** Orthotopic diversions can be made from small or large bowel. The key to obtaining a low-pressure reservoir with good volume is to detubularize the segment and reconfigure it into a spherical shape.

6. **d. Preservation of the uterus.** Late urinary retention ("hypercontinence") in women appears to be primarily due to posterior displacement of the pouch into the vagina resulting in a kinking at the urethral anastomosis. A number of maneuvers have been suggested such as sacrocolpopexy, but preservation of the uterus appears to be the most promising.
7. **b. Avoid excess dissection anterior to the urethra.** In preserving the urethra for an orthotopic bladder in males, one should be careful of the dorsal venous complex and avoid deep bites into the pelvic floor, especially anterior to the urethra where the rhabdosphincter is the most developed.
8. **d. Complete the cystectomy and neobladder and refer for adjuvant chemotherapy.** Patients with node-positive disease have a poor prognosis, but up to 30% may be long-term survivors, especially with adjuvant chemotherapy. Orthotopic diversion should not impact survival, and if a patient is highly motivated to avoid a stoma this option can still be pursued.
9. **b. Evaluation of renal function.** In order to consider a patient a candidate for a continent diversion, he/she must have a GFR in excess of 35 mL/min and his/her kidneys must be capable of concentrating and acidifying the urine.
10. **d. Pudendal nerve.** The rhabdosphincter is innervated by the pudendal somatic nerve. The contribution to continence from the pelvic autonomic plexus is uncertain, though some nonrandomized studies suggest that preserving the neurovascular bundle posterolateral to the prostate may improve continence.
11. **d. The staples can become a nidus for stone formation.** Metallic staples have a high association with subsequent stone formation, as was seen in the long-term experience with the hemi-Kock pouch. Recent efforts to perform intracorporeal neobladder using minimally invasive techniques have advocated using metal staplers to construct the pouch, but early results have suggested a high risk of stones.
12. **e. Systemic chemotherapy with or without external beam radiation.** Local recurrence of urothelial cancer will not usually impact the neobladder function. Rarely, direct invasion of the

reservoir will cause bleeding or outlet obstruction. Prognosis is poor, and local resection is rarely successful in eradicating the tumor, so primary treatment should be systemic chemotherapy and possibly radiation (which can be safely applied around a neobladder).

13. **c. Does not generally require treatment.** In patients with orthotopic bladders, approximately one quarter have asymptomatic bacteriuria. After the initial 6 months, it is rare for patients to have symptomatic infections or pyelonephritis, and bacteriuria does not require treatment.
14. **c. Check postvoid residual.** Patients who have a change in their continence after initial good function should be evaluated for possible urinary retention. This is often the first sign of incomplete emptying. Urodynamics are not necessary unless the pouch is made from colon, since ileal neobladders are very reliably low-pressure with good compliance.
15. **a. Teach the patient intermittent catheterization.** All patients who are considered for a continent diversion should be willing and capable of doing self-catheterization. The likelihood of needing self-catheterization is lower in men than in women, with reported rates of 10% to 40% in most series.
16. **d. Have often been underpowered or affected by selection bias.** Quality-of-life surveys have not shown one type of urinary diversion to be superior over another, though the vast majority of studies have serious methodological flaws. Obviously randomized studies in this area have been impossible. Most patients are reasonably well adapted socially, physically, and psychologically to their diversion. The key to this adaptation is appropriate and realistic preoperative education.
17. **d. Percutaneous drainage of the fluid collection.** If a patient has an undrained urine leak postoperatively, percutaneous drainage is the first step. Nephrostomy tubes can be placed if a large urine leak does not respond to percutaneous drainage with optimal catheter drainage. Open surgical repair should be avoided if possible because the complication rate is high and success in closing the leak in the face of acute postsurgical inflammation is low.

18. **c. Evaluate for possible vesicovaginal fistula.** A woman with persistent incontinence should be evaluated for a pouch vaginal fistula. This is most common when the anterior vaginal wall is removed with the specimen. It is best prevented by interposition of an omental pedicle but still can occur. Evaluation is easily performed with a speculum exam with methylene blue in the bladder.
19. **e. Renal ultrasound, chemistry panel and vitamin B12 level every 1 to 2 years.** Long-term risks in diversion patients include late ureteral stricture, stones, and vitamin B12 deficiency. All of these can be silent, so long-term routine follow-up is required. Most primary care physicians are not familiar with this.
20. **d. Continuous IV narcotics to optimize pain management.** New ERAS protocols have resulted in shorter hospital stay. Avoidance of bowel prep and the use of the μ opioid inhibitor alvimopan have been proven effective in randomized trials.
21. **b. Stenosis of the afferent nipple valve.** Afferent nipple stenosis is a well-documented late complication of the classic hemi-Kock pouch with an intussuscepted nipple valve antireflux mechanism. Treatment includes nephrostomy tube placement and endoscopic incision of the valve mechanism.
22. **d. The surgery can be performed through a smaller incision with lower blood loss.** A large series of patients undergoing robotic-assisted cystectomy with extracorporeal diversion showed no decrease in hospital stay or early or late complications compared to open series, and this was confirmed in one recent randomized trial from Memorial Sloan Kettering. Oncologic efficacy appears to be similar but long-term results are not yet available.
23. **e. Cystoscopy and laser lithotripsy of the stone.** Stones can occur in all types of continent diversions. These stones should be removed endoscopically while they are still small because they will inevitably grow and be more difficult to manage when larger.
24. **a. Reassurance that the nighttime continence will likely improve with more time.** Patients typically attain nighttime continence more slowly than daytime, with patients reporting

improvement out to 1 to 2 years. Early on there is an obligate nocturnal diuresis from the pouch that aggravates this. If the patient has good daytime control, further efforts to strengthen the sphincter are unlikely to help the nighttime continence. There is no role for anticholinergics or urodynamics in this setting.

Chapter Review

1. Urinary diversion has developed along three paths: conduit, continent cutaneous diversion, and most recently orthotopic diversion.
2. Long-term complications of the ileal and colon conduits include stomal stenosis, peristomal hernia, pyelonephritis, calculus formation, ureteral obstruction, and renal deterioration.
3. The major long-term problems with continent cutaneous diversion relate to function of the efferent continence mechanism, and surgical revision is often required.
4. The orthotopic neobladder allows for volitional voiding, avoids the need for an appliance and only require self-catheterization in a minority of patients.
5. 80%–90% of males and 75% of females undergoing cystectomy are potential candidates for orthotopic reconstruction.
6. Patient must have a healthy urethra and adequate external sphincter function to maintain continence.
7. The bowel segment used for the reservoir should be completely detubularized and reconstructed into a spherical shape.
8. Ultimate storage volume of the mature pouch should be at least 300–500 mL at low pressure.
9. The presence of CIS, multifocal tumor, or extravesical disease should not preclude orthotopic diversion if frozen section of the urethral margin is negative at surgery.
10. The most significant risk factor for a urethral tumor recurrence in men after orthotopic diversion is the presence of prostatic stromal invasion on final pathology.
11. The most important risk factor in women for urethral tumor involvement is bladder neck or anterior vaginal wall involvement

with cancer, best evaluated on bimanual examination under anesthesia.

12. Intraoperative frozen-section analysis of the urethral margin accurately identifies urethral involvement in both men and women. Preoperative biopsy of the prostatic urethra or bladder neck is not mandatory.
13. Local pelvic recurrence should occur in less than 10% of patients undergoing cystectomy and rarely interferes with the function of the neobladder.
14. Adequate renal function (eGFR exceeding 35 or 40 mL/min) is recommended for patients considering continent diversion.
15. Older age and obesity are not contraindications to orthotopic diversion.
16. Orthotopic diversion can be performed after prior pelvic irradiation or radical prostatectomy in carefully selected male and female patients.
17. In patients with prior bowel resection the prior anastomosis should be taken down and that segment used for the orthotopic diversion rather than choosing a new site, to avoid devascularization of the bowel.
18. The striated rhabdosphincter muscle fibers and associated innervation from the pudendal nerve are concentrated in the area anterior and lateral to the proximal urethra in both males and females.
19. In male patients, one should obtain careful control of the dorsal venous complex, and avoidance of deep suture bites into the pelvic floor muscles is critical.
20. In female patients, the endopelvic fascia and levator muscles should not be disturbed.
21. Reservoirs made from ileum or combined ileum and colon appear to have the best physiologic properties for orthotopic diversion.
22. Isolation of the segment of bowel to be used for the diversion must be performed carefully to preserve blood supply to the pouch and bowel anastomosis.
23. The addition of an antireflux mechanism does not appear to be necessary for preservation of the upper tracts and prevention of

infections, at least in the intermediate term.

24. Daytime continence develops gradually over 3–6 months in most patients and is ultimately achieved in 80%–90% of both male and female patients.
25. Persistent nocturnal incontinence is common, observed in 20%–50% of patients. Nocturnal continence may continue to improve beyond 12 months from surgery and may improve with timed voiding.
26. Factors influencing continence rates include age, intestinal segment used, and application of a prostate-sparing technique.
27. Failure to empty or urinary retention has been reported in 4%–10% of men and 20%–60% of women.
28. Early morbidity and mortality of cystectomy and orthotopic diversion are not increased compared with ileal conduit.
29. Undrained urine leak is best managed using percutaneous drains and bilateral nephrostomy tubes when necessary. Early open surgical revision should be avoided.
30. The majority of late urologic complications related to the neobladder can be managed using endoscopic techniques.
31. Patients require regular follow-up imaging to identify upper tract obstruction and stones, which may be clinically silent.
32. Follow-up should include physical and pelvic examination and urethral cytology to identify urethral recurrence, which occurs in approximately 10% of males and rarely in females.
33. Most quality-of-life studies that have evaluated and compared patients undergoing various forms of urinary diversion have been criticized for methodologic problems that limit their conclusions.
34. The current body of published literature is insufficient to conclude that any form of urinary diversion is superior to another on the basis of health-related quality-of-life outcomes, though most studies report better body image with neobladders.